ELECTRON MICROSCOPY FACILITY

C-WING, GROUND FLOOR, GC-04, DIVISION OF BIOLOGICAL SCIENCES
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ULTRAMICROTOME SAMPLE PREPARATION & SECTIONING

Workstation Location: Room GC-04 (Sample Prep Area)	**Equipment:** Leica Ultramicrotome
Pre-requisite: Users must bring pre-cured/embedded resin blocks	**Target Thickness:** TEM (60–90 nm) / Semithin (0.5–1 µm)

PRECAUTION & WORKBENCH GUIDELINES: Users must wear appropriate PPE (gloves, shoe covers, lab coat). Glass and diamond knives are extremely sharp—handle with care. Do not remove tweezers or station accessories from GC-04 to prevent cross-contamination.

SECTIONING PROTOCOL

1. **Resin Block Trimming:** Mount the pre-polymerized resin block into the specimen holder. Use a fresh razor blade to trim excess resin, forming a trapezoidal/rectangular block face (~0.5 × 0.5 mm) with parallel top and bottom edges.
2. **Knife Preparation:** Mount a glass knife or authorized diamond knife onto the stage. Fill the attached water boat with Milli-Q water until a flat/concave meniscus forms at the cutting edge.
3. **Block Alignment:** Set the clearance angle (4°–6°). Align the resin block face manually to ensure it is perfectly parallel to the knife edge under binocular illumination.
4. **Sectioning & Thickness Check:** Set the sectioning speed (1.0–2.0 mm/s) and thickness. Monitor section interference colors in the water boat:
    - **Silver** / **Light Gold (~60–70 nm):** Optimal for TEM imaging.
    - **Gold (~80–100 nm)** / **Blue-Green (>150 nm):** Suitable for semithin optical analysis.
5. **Grid Collection:** Submerge a clean, glow-discharged EM grid (200–300 mesh) into the water boat and gently collect the floating ribbon. Blot excess water with filter paper edge and air-dry for at least 10 minutes.
6. **Cleaning & Logbook:** Drain the knife boat. Clean diamond knives strictly using approved Styrofoam cleaning sticks (never use paper/cotton). Wipe down the stage with 70% ethanol, return all tools, and sign the facility logbook before leaving.

Note: Ultramicrotomy is designed to yield ultrathin, uniform sections of embedded biological specimens without introducing structural artifacts. Adherence to blade handling, precise block trimming, and boat water-level management is required to achieve high-quality TEM grid preparation.